Economics S5040: Dynamic Stochastic Modeling
Lecture 16 — Computing the Income Fluctuation Problem

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Central question: Once we have formulated the household's self-insurance problem, how do we compute and interpret its optimal policy?

Plan for an 80-minute class

Minutes	Topic
0–10	Review the IFP and Euler inequality
10–30	Discretization and value function iteration
30–45	Consumption and saving policy functions
45–58	Simulation: income, assets, and consumption
58–72	Euler/time iteration and the idea of EGM
72–80	Compare methods and economic conclusions

1 Review

The household solves

$$V(a,z) = \max_{a' \geq \underline{a}} \left\{ u(Ra + y(z) - a') + \beta \sum_{z'} P(z,z') V(a',z') \right\}.$$

The state is (a,z) and the policy is

$$a' = g(a,z).$$

Consumption follows from

$$c(a,z) = Ra + y(z) - g(a,z).$$

2 Discrete value function iteration

Choose an asset grid

$$\mathcal{A} = \{a_1, \ldots, a_N\}$$

and a finite income-state space

$$\mathcal{Z} = \{z_1, \ldots, z_M\}.$$

Given a current guess V, define

$$(TV)(a_i, z_j) = \max_{a'_k \in \mathcal{A}} \left\{ u(Ra_i + y(z_j) - a'_k) + \beta \sum_{\ell=1}^{M} P_{j\ell} V(a'_k, z_\ell) \right\},$$

where infeasible choices with nonpositive consumption are excluded.

The algorithm is familiar:

1. Guess V_0.
2. Compute $V_{n+1} = TV_n$.
3. Record the maximizing asset choice at every (a, z).
4. Stop when $\|V_{n+1} - V_n\|_\infty$ is small.

3 What should the policy functions look like?

Plot both

$$a' = g(a, z)$$

and

$$c = Ra + y(z) - g(a, z)$$

against current assets for each income state.

Economic questions matter more than the picture itself:

- Where does the borrowing constraint bind?
- Does consumption rise one-for-one with current resources?
- When income is temporarily low, does the household run down assets?
- When income is high, does it rebuild its buffer stock?
- How does persistence of income risk change desired assets?

4 Simulation turns a policy into economic behavior

Given $g(a, z)$ and a simulated Markov path z_t, generate

$$a_{t+1} = g(a_t, z_t),$$

$$c_t = Ra_t + y(z_t) - a_{t+1}.$$

Plot y_t, c_t, and a_t.

The central empirical-looking implication is that consumption should generally fluctuate less than labor income because the household uses assets to absorb some of the income variation.

5 Why look beyond VFI?

VFI is robust, but it repeatedly solves a maximization problem at every grid point. The Euler equation contains additional structure:

$$u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]$$

when the borrowing constraint does not bind.

A policy-iteration or time-iteration method works directly with this condition. Instead of iterating on the value function, we iterate on a consumption policy.

6 The idea behind endogenous grid methods

Suppose we choose a grid for next-period assets or saving rather than current assets. For an interior point, the Euler equation tells us current consumption from expected future marginal utility. Once c and saving are known, the budget constraint tells us which current asset level would make that choice feasible.

Schematically:

$$\boxed{\text{choose saving grid}} \longrightarrow \boxed{\text{Euler equation gives } c} \longrightarrow \boxed{\text{budget constraint gives current assets}}.$$

That last grid is generated by the economics of the problem—hence *endogenous grid method*.

The method avoids a costly numerical maximization at every exogenous asset-grid point. The price of this speed is that it uses more of the model's special structure.

7 The constraint still matters

The Euler equality applies only to interior saving choices. At the borrowing boundary, the policy must respect

$$a' \geq \underline{a}.$$

A numerical method therefore needs explicit treatment of the constrained region. This is not merely a coding issue: the kink in the policy reflects a genuine economic restriction on consumption smoothing.

8 Comparing methods

Method	Main object	Main advantage
VFI	value function	transparent and robust
Policy/time iteration	policy / Euler equation	faster when structure is usable
EGM	policy on endogenous grid	avoids repeated maximization

The economic solution should be the same. The methods differ in how efficiently they exploit the model's structure.

9 A useful comparative-static experiment

Hold mean income fixed and increase income risk. Re-solve the model and compare

$$g_{\text{low risk}}(a, z) \qquad \text{and} \qquad g_{\text{high risk}}(a, z).$$

This gives a computational measure of precautionary saving. It also connects directly to the two-period experiment from Lecture 15.

A second experiment is to make low income more persistent. The household then expects a low-income realization to last longer, changing the value of the buffer stock.

End-of-class summary

VFI gives us a robust benchmark solution.
Policy functions turn the Bellman equation into observable behavior.
Simulation shows consumption smoothing and self-insurance.
Euler-based methods and EGM exploit structure to solve faster.

Readings

- `https://python.quantecon.org/ifp_discrete.html`
- `https://python.quantecon.org/ifp_opi.html`
- `https://python.quantecon.org/ifp_egm.html`